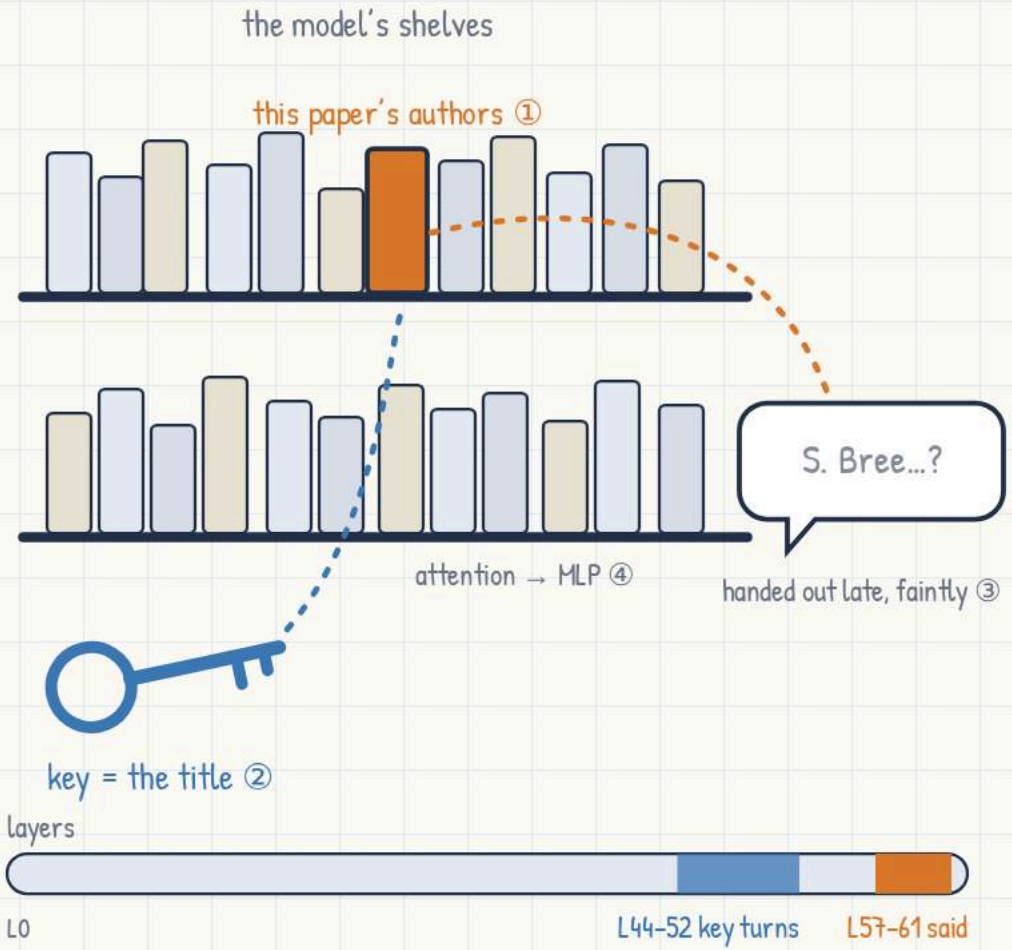
Keys and shelves

Is the book on the shelf, or is the key failing to get it out?



① Is the book on the shelf? Yes

True title beats false titles (≈0.44 at L62(§6.9)); a probe on the title token reads paper-specific authors from L24 (re-checked)

Lines A, B · Stage 5, 7.3, 8D, 10, 11

② Key = the title, read into the name positions at L44–52

Whole prompt: 0.82 at L48, ≈1.0 from L52; name positions alone 0.61 → 0.85; ':' alone ≤ 0.17(E1, P1; 1,186 papers); knockout after L52 costs almost nothing

Line C · Stage 11, 12 P1

③ Book there but stuck? Not supported

Target – control ≈ 0 at L24–L40, slightly negative at L48 (−0.04 to −0.05), grows only at L57–61 (+0.21 at output(§6.4)): retrieval itself is weak

Lines A, C · Stage 7.2, 10

④ Which shelf? Both, in order

Attention brings the title in at L48 (0.23); MLP takes over at L52–56 (0.18, 0.22)(P2, 1,186 papers). One-layer patching is a lower bound; whether the MLP step is memory lookup is not yet separable

Stage 12 P2

Caveat: a readout not finding it ≠ absent. Stage 13: ':' and name positions hold the task; the title position holds paper attributes and meaning beyond the words, not shown to be paper identity.

How we read the model



Score = rank of the correct next name token,

as log10(1 / rank): 0 = rank 1, -1 = rank 10

7 title conditions: the true title, 5 matched false titles, no title

title-specific signal

= true title - mean(5 false titles)

Same name, same tokens: what is left is knowing

this paper

Lenses (model weights never change; Jacobians to L63 averaged over prompts): public = 557 WikiText-103 prompts (Neuronpedia) · self100 = 100 frozen WikiText-103 prompts · fitA = 100 other WikiText-103 prompts (Stage 9: one skipped prompt replaced -> 100 valid) (n_prompts: 351_stage10/config/lenses.json; Stage 9 final_status.json)

Same papers, two ways of grouping

Stages 1–9 compare A vs B; Stages 10–11 compare target vs control. Counts from the 6,841-paper cohort.(Stage 1, 351_audit.json group_counts)

	recognised (target)	not recognised (control)
R0 · no true author written	A = 4,110	B = 2,186
R1 · some authors written (HR2 > 0.5)	C = 511	D = 34
Total	**target = A + C = 4,621**	**control = B + D = 2,220**

Stages 1–9

A vs B: both R0, so the contrast is purely "knows but cannot say"

Stages 10–11

(A + C) vs (B + D): adds 511(351_audit.json) partial-recall papers, almost all on the target side

Check done: with R0 only (A vs B) the output gap is +0.199, versus +0.207 for target vs control(Stage 10, final L63, within-bin; §6.13), and every layer looks the same. The two phases compare like with like.

How groups and contrasts are built

labels: group lead's pipeline on the same Qwen3-32B · we use them unchanged

open answer -> R0 / R1 / R2

greedy, chat template

fuzzy name match: M, H, U

HR2 = (H+U) / (M+H+U)

R2: HR2 <= 0.5 · R0: M = 0 · R1: M > 0

a fixed rule, no human or model judge

multiple choice -> target / control

true list + 3 real teams of other papers

(same field, team size, citation bin, ~year)

4 orders · p(letter) from next token

target: 4/4 and mean p(true) >= 0.5

control: <= 1/4

5 false titles per paper

same field, own year kept

min-cost on |dyear|/5 + |dauthors|

+ |dtitle tokens| + 1.5 |dcitation bin|

no shared surname, Jaccard <= 0.6

79% strict tier (Stage 10, 35,898 of 45,540 pairs, conditions_manifest.json)

statistics

target vs control only within citation bin

10 bins with >= 30 per group (Stage 10, §6.2)

weight $n_t n_c / (n_t + n_c)$

95% bootstrap over papers, 2,000

sensitivity: drop bins 0-4

Can we trust the numbers?

fast = slow

One forward pass vs one per prefix: max gap 0.37 logit (bf16 rounding); ranks 1-41 identical (Stage 10, 12 contexts, parity.json)

matches Stage 6

972 tokens: same token ids, median log-prob gap ~1e-5 (Stage 10, crosscheck_stage6.csv)

negative control

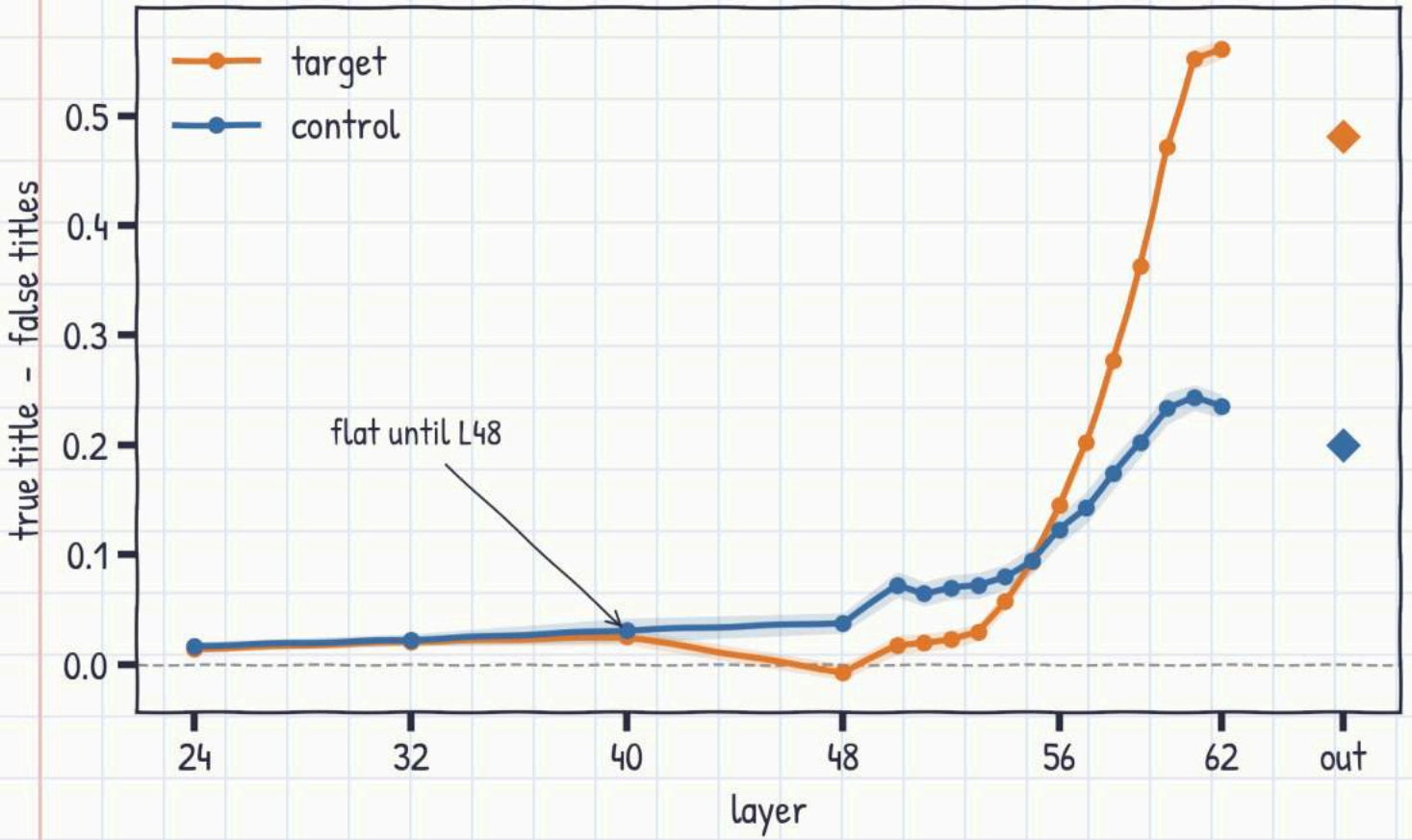
Punctuation tokens show ~0 title-specific signal at every layer (Stage 10, max 0.014, 9,108 papers, §6.5)

Caveat: the parity tolerance was raised 0.25 -> 0.5 after seeing these bf16 gaps. (parity.json tolerance_amendment)

Stage 10

Knowing the paper shows up only at the end

title-specific signal · public Jacobian lens (lines) and model output (diamonds)



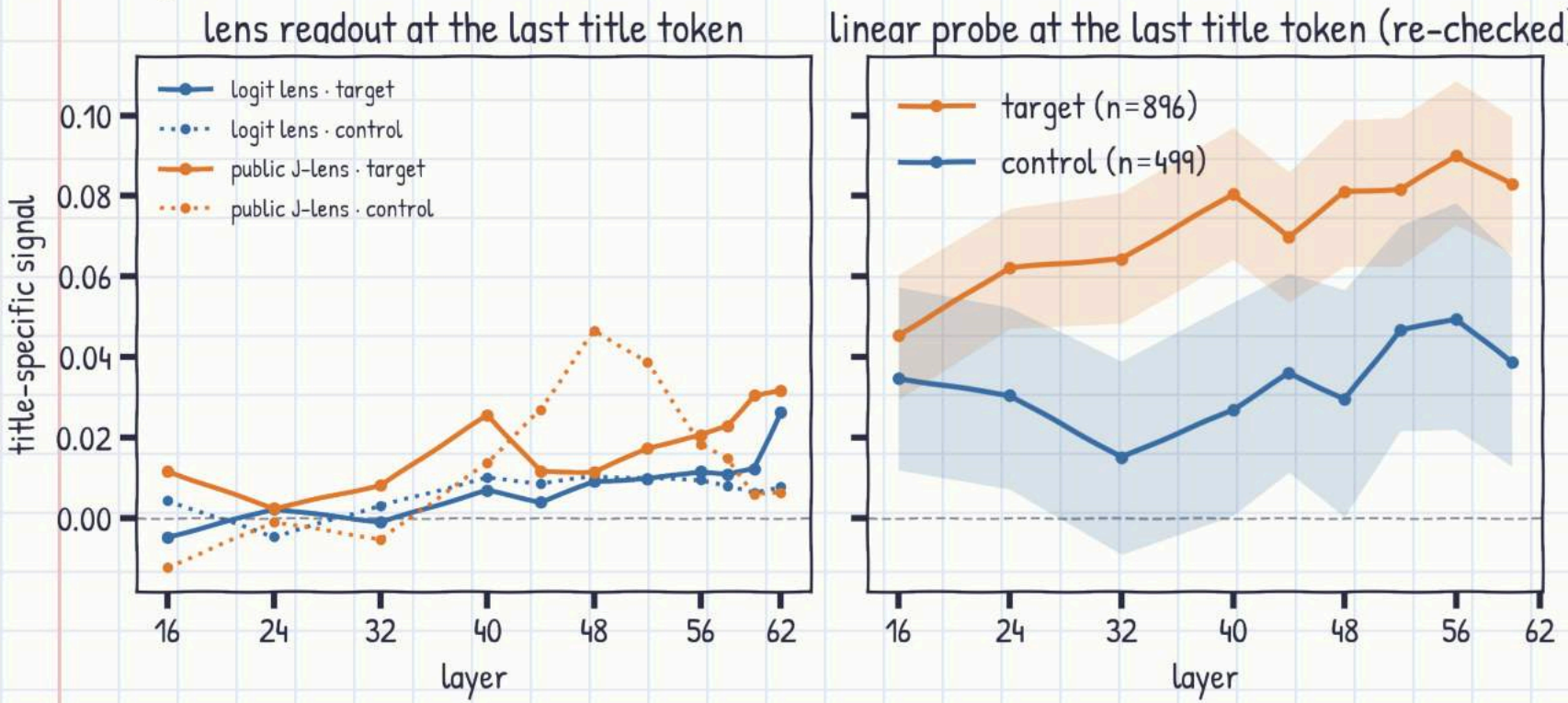
L24–L48: ~0 for both groups (Stage 10, all readouts max 0.04, §6.3)

rises at **L56**

L62: **target +0.56** vs **control +0.24**
(Stage 10, public lens L62, 4,621 + 2,220 papers, §6.3 fig.)

Read early: a probe finds it at the title

title-specific signal at the last title token · lenses: all 9,108 papers · probe: re-checked, title reuse removed: 1,863 papers (896 target, 499 control) (Stage 11 E4; Stage 13 E4b)



lenses: **~0.01–0.03** (Stage 11 E4, 4,621 + 2,220, §6.11)

probe: **0.05 -> 0.08** (Stage 13 E4b, 1,863 papers, first name token only)

target > control by **+0.03–0.08** (E4b, 896 vs 499, within bin)

re-checked: team folds, title reuse removed

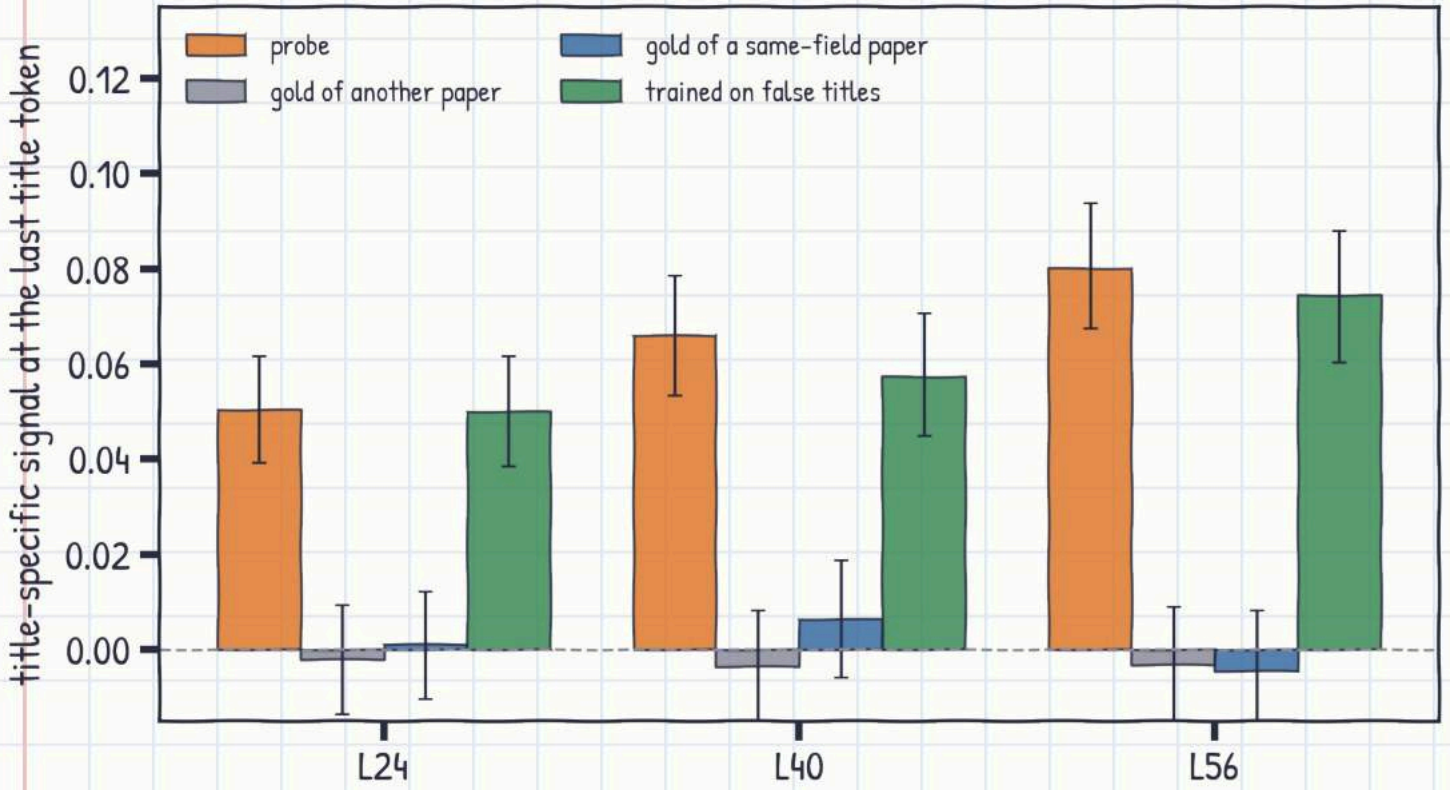
scores the first name token only (mostly an initial); subtracts false_0 only

read early (title, from L24), **spoken late** (names, L57–61)

Probe scores only the first name token (mostly an initial); all-token, by-role version (addendum 9, done): surname first token L40 +0.071 [0.058, 0.085] (1,395 papers; e4b_alltok.md).

The probe signal is paper-specific

last title token · 5-fold ridge probe · re-checked: author-team folds, title reuse removed



another paper's authors: **0**

same-field paper: **0** (Stage 12 X1 + Stage 13 re-check, §6.12)

so it is **this paper**, already at L24

lenses cannot read it

Probe scores only the first name token (mostly an initial); all-token, by-role version (addendum 9, done): surname first token L40 +0.071 [0.058, 0.085] (1,395 papers; e4b_alltok.md).

Stage 11: intervene, do not just read

593 + 593 papers: min(60, n_t, n_c) per citation bin, seed 11 (Stage 11 sample.parquet, §6.11)

patch uses false_0 · knockout masks all heads · all 9,108 for probe and chain (Stage 10 manifest.json)

patch

copy the true-title hidden state into the false-title run at one layer

knock out

block attention to the title tokens (all heads) from layer l to the last layer

probe

linear map from the last title token to the output state

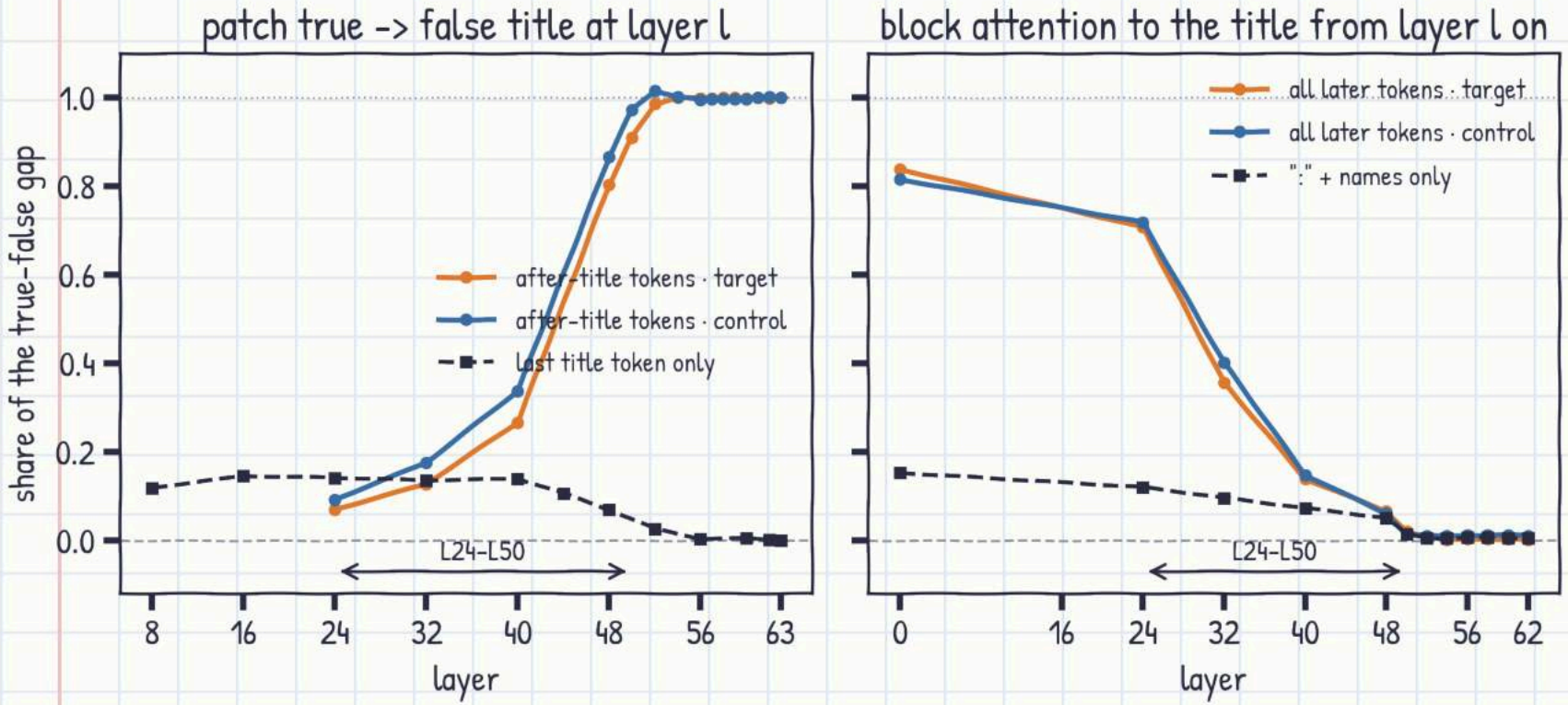
chain

"Authors: A1, A2, ..." instead of each author alone

Score: rank of the correct name token at the model output; "share of the true-false gap" = (intervened - false) / (true - false).

The title is read in L24-L50, early

model output · share of the true-false gap · CIs about +/-0.02 (Stage 11 E1/E2, 593 + 593 papers, §6.11)



L52+: title no longer needed

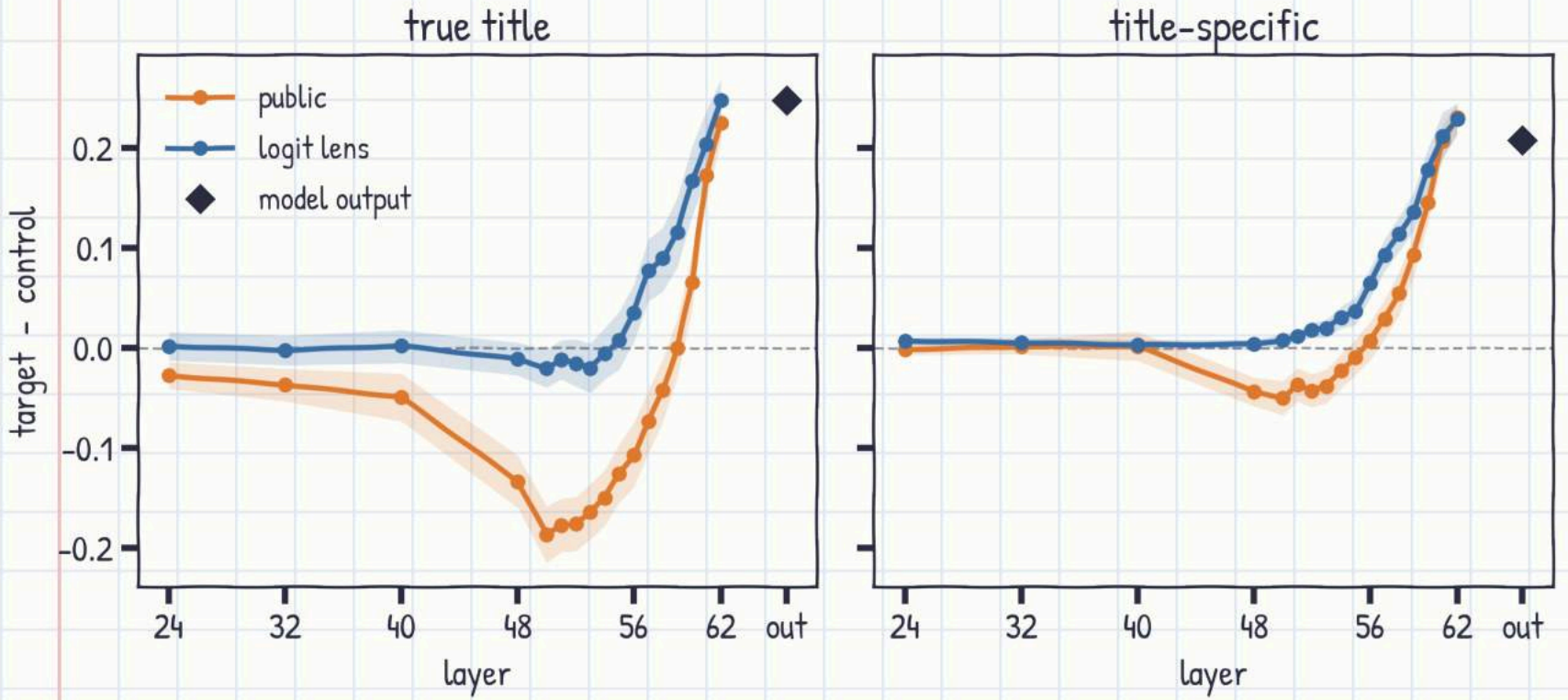
the information enters the **name positions at L44-52**, partly via the tokens right after the title

same timing for target and control

Stage 10

Target pulls ahead only in the last layers

target − control, within citation bin, 95% CI



Middle layers: **no head start**

L62 / output:

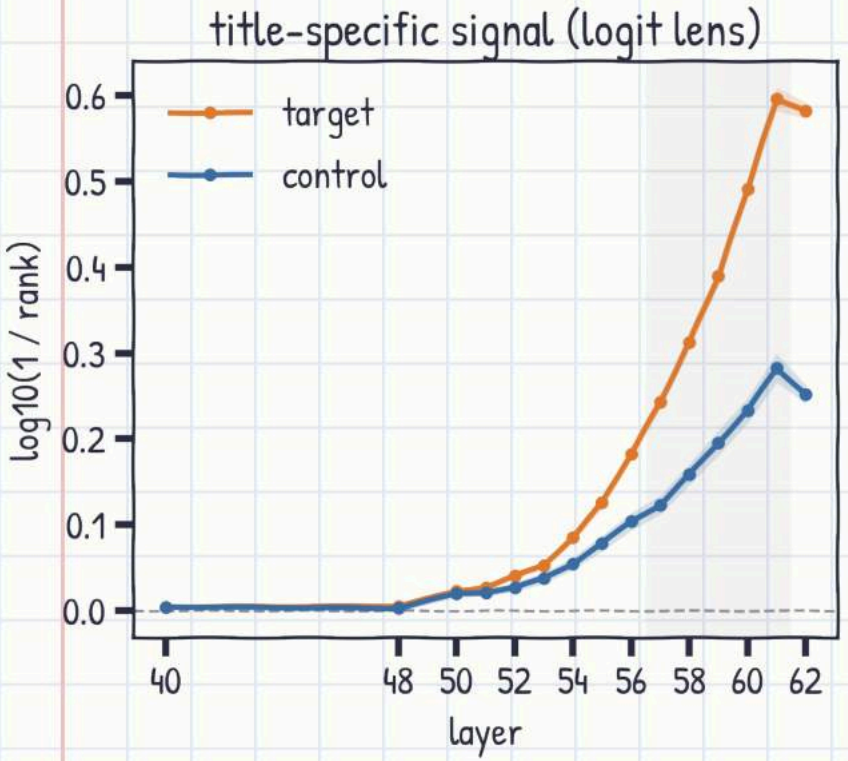
+0.21 to +0.23 (Stage 10, L62 + final L63, 3,734 + 2,182 papers, §6.4)

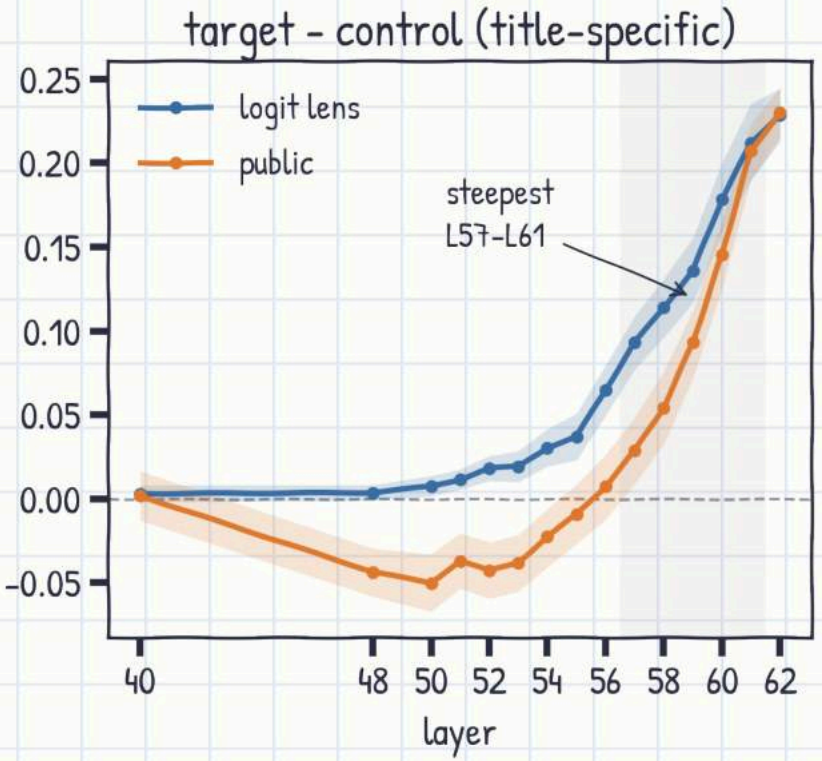
almost all title-specific

R0 only (no author written): output +0.199 vs +0.207 (Stage 10, 3,423 + 2,151 R0 papers, §6.13)

It builds up over the last dozen layers

fine layers L40–L62 · title-specific signal · 95% CI





Logit lens: target - control

+0.007 at L50

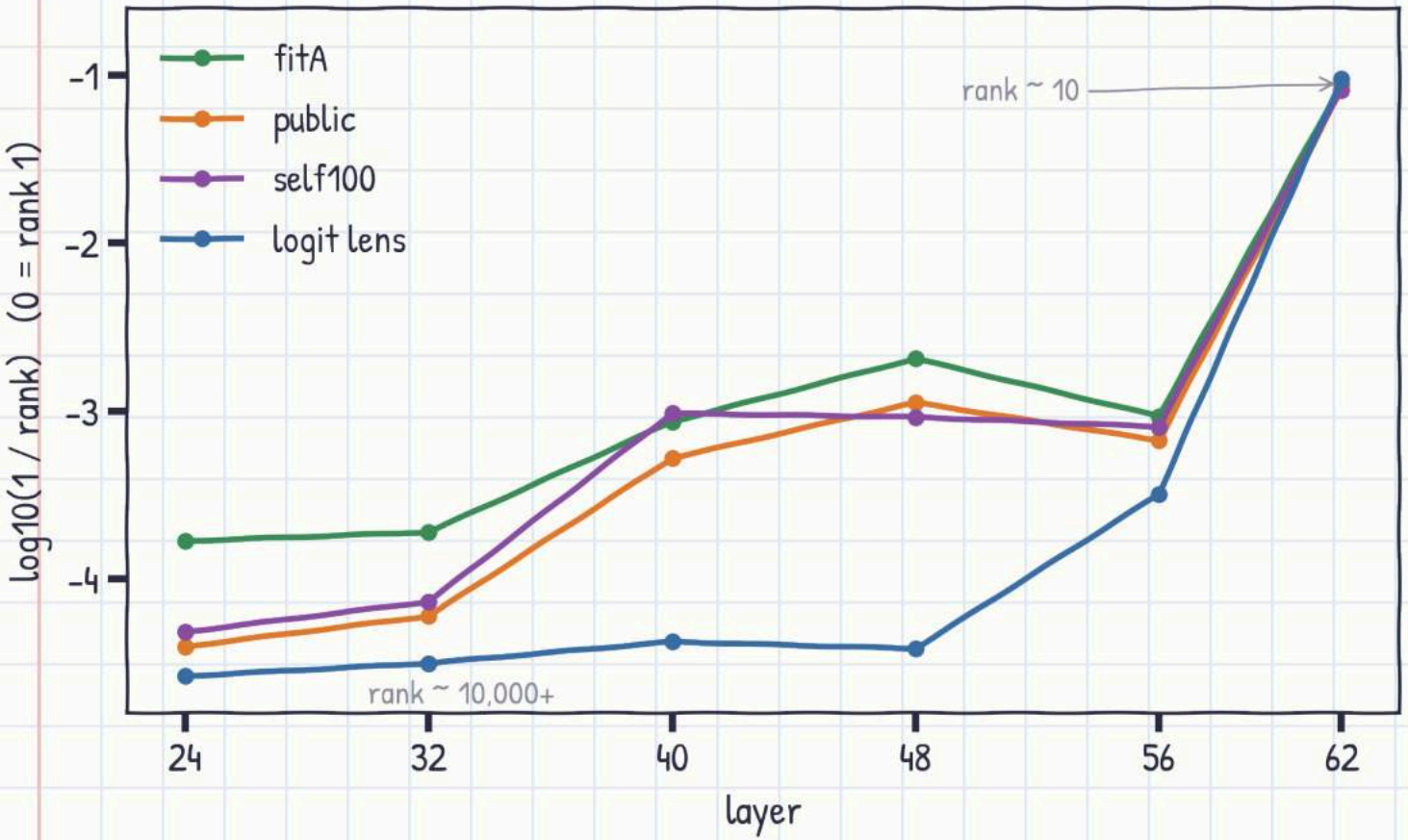
+0.23 at L62 (Stage 10 logit lens, 3,734 + 2,182 papers, §6.10)

steepest rise **L57–L61**

Stage 10

Names become readable late

all 9,108 papers · true title (Stage 10 cohort, §6.1)



Logit lens: almost nothing until **L48**

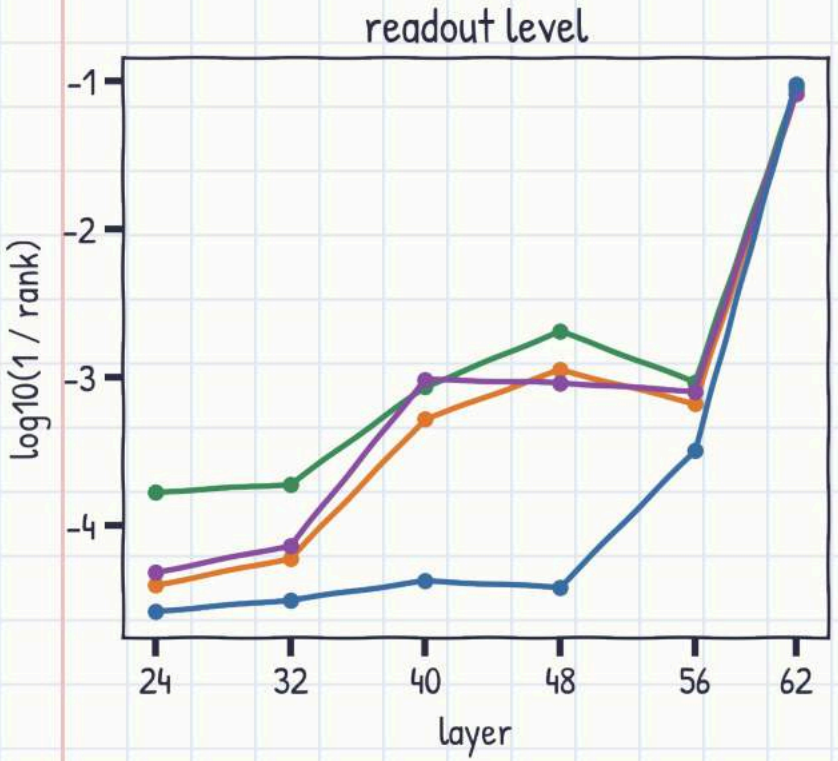
Jacobian lenses read more from **L40**

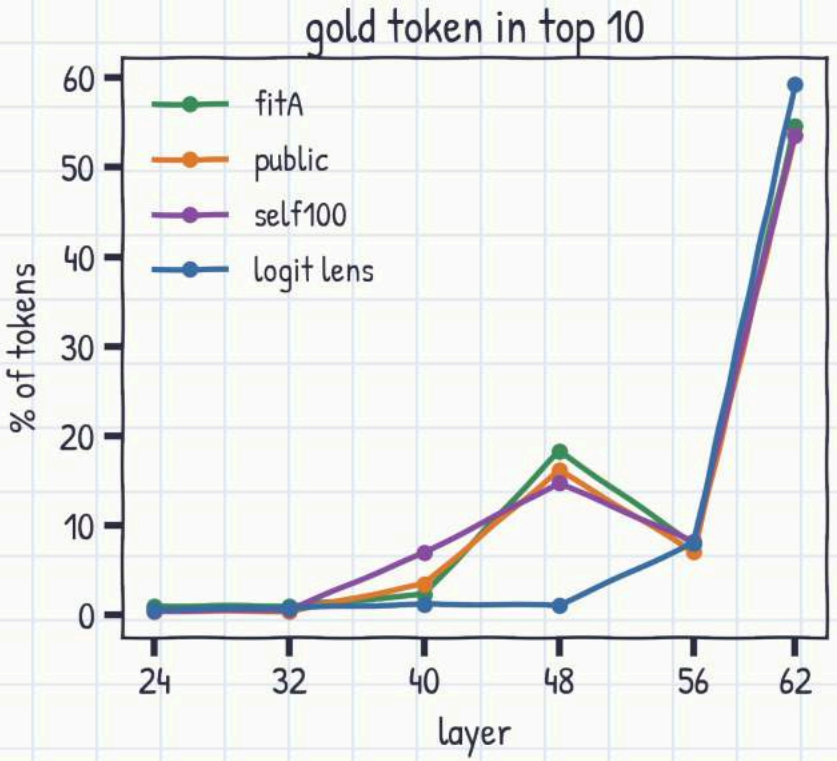
Everything jumps at **L62** (rank ~10) (Stage 10, logit L62 -0.93, §6.1)

Stage 10

Which lens reads best?

all papers · true title





Middle layers:

fitA best; at L48 15–18% top-10 vs 1% for logit

(Stage 10 X3, 9,108 papers, §6.9)

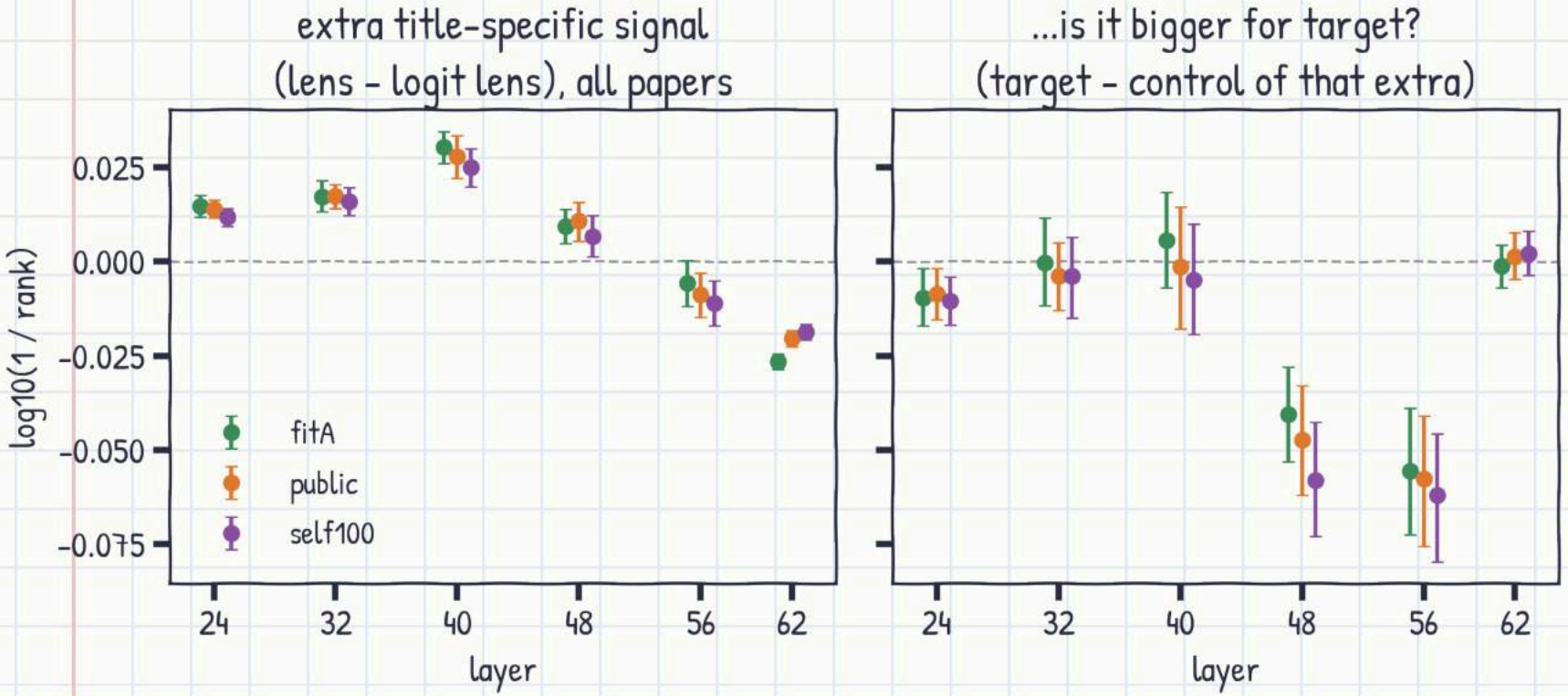
L62: logit lens best (59%)

(same, L62)

Stage 10

Do lenses reveal hidden paper knowledge?

paired differences, 95% CI



Extra signal: **+0.01 to +0.03** (Stage 10 X3, L24–L48, 9,108 papers, §6.9)

vs ~0.44 at L62 (same, L62)

Not bigger for target
(even negative at L48–56)

Hypothesis	M1: top word of the lens in middle layers	M2: probe reads non-name attributes	M3: change-of-task control	Current result (2026-11-05)
H1 task / format: the middle layers represent "output an author list"	mostly "author", punctuation, common name starts; unrelated to the title	paper attributes not readable or very weak	the same title under different questions separates early in the middle layers	see the M3 and X1 results
H2 paper identity: the middle layers represent "which paper this is"	title-related words (copied from the title); change with a false title	field, year (inferred from the title), citations and recognition all readable, and earlier than the names	the same paper under different questions is close in the middle layers and separates by task only late **Result:** not above the L4 text level, not supported	**M3 does not support it; X3 shows a semantic representation beyond wording in the middle layers, but it cannot be separated from paper identity; H2 currently rests on M2 and X1, and M2's Qwen-over-BERT advantage is reference only (addendum 2).** Also, the earliest decodable order in X4 is first author (L8) before year (L40), opposite to the H2 prediction. **X2 does not support it.**
H3 author information already there, just not in vocabulary directions	not necessarily	the authors' first token readable by the probe, not by the lens	an author-trained probe transfers across tasks; an author-specific lens (M4) reads names in the middle layers	**Text baseline not weaker than Qwen, a text explanation cannot be excluded: insufficient evidence** (X4: on the given-name first token BERT +0.088 ≥ Qwen L8 +0.018; the L8 signal sits mainly on the surname, see X4); **X2 passes by rule but the subspace decodes nothing: still insufficient**

Three hypotheses and their predictions

Hypothesis	M1: top word of the lens in middle layers	M2: probe reads non-name attributes	M3: change-of-task control	Current result (2026-11-05)
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Overall (2026-11-03): H2 partly supported but not established; H3 insufficient evidence. H2 support comes from M2 (paper attributes readable at the title) and X1 (replacement effect independent of the question); M3 does not support it, X3 cannot separate meaning from paper identity, the X4 order is opposite to the H2 prediction, X2 does not support it (2026-11-05). H3: X2 passes by rule, but the author subspace hardly decodes authors, so the evidence is still insufficient.

English translation of the report table (Stage 1–13 report v28/v29, part 4 "三种假设及其预测"); numbers unchanged. Sources: 381_stage13/reports/m3_report.md, x1_report.md, x3_report.md, x4_report.md, x4_textbase2.md, x2_report.md.

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